BTS-VIII(S/S)-09-24-2913 Reg. No. | | | | | | | | |

A

B.Tech. Degree VIII Semester Special Supplementary Examination September 2024

ME 19-205-0810 OPERATIONS RESEARCH

(2019 Scheme)

Time: 3 Hours Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Solve problems using optimisation methods.
CO2: Develop mathematical skills to approach real life industrial problems.
CO3: Inculcate ideas to solve problems on Transportation and Queueing models.
CO4: Import and analyse case studies to solve future problems.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

		$(8 \times 3 = 24)$	Marks	BL	CO	PO
I.	(a)	What are the different phases of OR study?	3	2	1	1
	(b)	How does graphical method give indication that the solution is unbounded?	3	1	1	2
	(c)	What is degeneracy? How do you avoid cycling in Simplex method?	3	2	2	2
	(d)	Define duality in linear programming with the help of an example.	3	2	2	1
	(e)	How do you solve a degenerate transportation problem?	3	1	3	2
	(f)	How do you justify that an Assignment Problem is a special type of Linear Programming Problem?	3	1	3	2
	(g)	What is forward and backward recursion in dynamic programming?	3	2	2	2
	(h)	Briefly explain birth and death process in queueing theory.	3	2	2	2

PART B

$(4 \times 12 = 48)$

		Marks	BL	CO	PO
II.	A company is interested in the analysis of two products which can be made from the idle time of labour, machine and investment possible in this regard. It was found on investigation that the labour requirement for the first and second products was 2 and 3 units respectively and the total available man hours was 24. Only product 1 requires machine hour utilisation of one hour per unit and at present only 9 spare machines are available. Product 2 requires one unit of a byproduct per unit and the daily availability of the byproduct is 6 units. According to the marketing department, the sales potential of product 1 cannot exceed 5 units. In a competitive market, product 1 can be sold at a profit of ₹3 and product 2 at a profit of ₹5 per unit. Formulate the problem as an LPP. Determine graphically the feasible region. Identify the redundant constraint.	12	L3	2	2

OR

(P.T.O.)

			Marks	BL	CO	PO
III.	(a)	Explain briefly a Convex set and a Nonconvex set in linear programming.	4	L2	1	2
	(b)	Consider the following LP Maximise $z = 2x_1 + 3x_2$. Subject to $x_1 + 3x_2 \le 6$ $3x_1 + 2x_2 \le 6$ $x_1, x_2 \ge 0$. Show how the infeasible basic solutions are represented on the graphical solution space.	8	L3	1	2
IV.	(a)	In simplex method, how will you identify that the solution is unbounded ?	3	L2	1	1
	(b)	Solve the following problem by two phase method Minimise $z = 8x_1 - 2x_2$ Subject to $-4x_1 + 2x_2 \le 1$ $5x_1 - 4x_2 \le 3$ $x_1, x_2 \ge 0$.	9	L3	1	2

OR

V.		Solve the following problem by two phase method. Minimise $z = 4x_1 + x_2$ Subject to $3x_1 + x_2 = 3$ $4x_1 + 3x_2 \ge 6$ $x_1 + 2x_2 \le 4$ $x_1, x_2 \ge 0$	12	L3	1	2
VI.		Solve the following transportation problem.	12	L3	3	2

Plant/Ware house	W1	W2	W3	Available
A	25	17	25	300
B	15	10	18	500
Requirement	300	300	500	

OR

VII.		A job shop has 4 men A, B, C, D available for work on four separate jobs 1, 2, 3, 4. Only one man can work on any one job. The cost of assigning each man to each job is given in the following table. Find the optimal assignment such that the total cost of assignment is a minimum.	12	L3	3	2

Men/Jobs	1	2	3	4
A	20	25	22	28
B	15	18	23	17
C	19	17	21	24
D	25	23	24	24

(Continued)

			Marks	BL	CO	PO
VIII.	(a)	Explain your understanding about the relationship between the arrival rate and the average inter-arrival time.	4	L2	3	1
	(b)	In Rajasthan, babies are born at the rate of 1 birth in every 12 minutes. The time between births follows exponential distribution. Find the following. (i) The average number of births per year. (ii) The probability that the no births will occur in any one day. (iii) The probability of issuing 50 birth certificates in 3 hours given that 40 certificates were issued during the first 2 hours of the 3 hour period.	8	L5	3	2
		OR				
IX.		Lumiere travel arranges 1 week tours to southern Egypt. The agency is contracted to provide tourist groups with 7, 4, 7 and 8 rental cars over the next 4 weeks respectively. Lumiere travels subcontracts with a local car dealer to supply rental needs. The dealer charges a rental fee of $220 per car per week plus a flat fee of $500 for any rental transaction. Lumiere, however, may elect not to return the rental cars at the end of the week, in which case the agency will be responsible only for the weekly rental. What is the best way for Lumiere travel to handle the rental situation? Use dynamic programming to find the solution.	12	L5	2	2

Bloom's Taxonomy Levels
L1 – 7.5%, L2 – 21.7%, L3 – 54.1%, L5 – 16.7%.

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